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Cogs 209 Mini-Project #1

Predicting AI Tool Adoption from Socioeconomic Status:
Is Artificial Intelligence Becoming a New Axis of Privilege?

1. Introduction and Hypotheses

AI tools like ChatGPT, Gemini, and GitHub Copilot are increasingly used for writing, coding, research, and professional tasks. As these tools grow in influence, a critical question emerges: is adoption distributed equitably across society, or is it becoming yet another marker of socioeconomic privilege? This project investigates whether income and education level are significant predictors of AI chatbot adoption among U.S. adults, and how much each factor independently contributes to that prediction.

Controlled experiments have found that workers given access to AI complete tasks significantly faster and at higher quality than those without (Noy & Zhang, 2023; Brynjolfsson et al., 2023). PwC's 2025 Global Workforce Survey of nearly 50,000 workers found that daily AI users are far more likely to report higher pay and productivity than infrequent users, and that workers with AI skills earn 56% more than peers in the same role without them (PwC, 2025a; 2025b). If adoption of these tools is already concentrated among higher-income and higher-educated groups, AI may compound existing socioeconomic inequality rather than democratize opportunity. Understanding who is and is not adopting AI is therefore relevant for policymakers, educators, and researchers concerned with equitable technology access.

We propose two hypotheses. First, that both income and education will be independently and significantly associated with AI chatbot adoption after controlling for age, gender, and race, and that education will emerge as the stronger predictor given that AI familiarity is largely mediated through professional and academic contexts. Second, that income and education will interact such that their combined effect on adoption exceeds what either variable predicts independently. Individuals who are both high-income and highly educated will show disproportionately higher adoption rates than those who are high on only one dimension, reflecting a compounding privilege effect.

2. Methods

2a. Data

We use Wave 152 of the Pew Research Center's American Trends Panel (ATP), a nationally representative survey of 5,410 U.S. adults conducted August 12–18, 2024. Data were collected online and by live telephone in English and Spanish using address-based sampling, with an oversample of Hispanic men, non-Hispanic Black men, and non-Hispanic Asian adults weighted back to population proportions. The dataset is publicly available at Pew Research Center Datasets following free account registration. After removing respondents with missing values on any key modeling variable, the analytic sample is n = 3,798.

The primary outcome is CHATUSE_W152, recoded as a binary indicator: 1 = has ever used an AI chatbot (ChatGPT, Gemini, or Copilot); 0 = has not. The primary predictors are education (educ_6cat: 1 = less than HS through 6 = postgraduate) and family income (inc_9cat: 1 = under $30k through 9 = $130k+), both treated as ordinal integers given their meaningful rank ordering. Demographic controls include age category (age_cat: 1 = 18–29, 2 = 30–49, 3 = 50–64, 4 = 65+), gender (reference: men), and race/ethnicity (reference: White non-Hispanic). Gender and race are dummy-coded using drop-first encoding to avoid perfect multicollinearity. Survey weights (WEIGHT_W152) were not applied in the predictive modeling comparison, as cross-validated accuracy comparisons are not affected by weighting; this is noted as a limitation for population-level inference.

2b. Computational and Statistical Methods

All analyses were conducted in Python 3.11 using scikit-learn (version 1.8), pandas, scipy, seaborn, and matplotlib.

Primary Classifier. The primary outcome, whether a respondent had ever used an AI chatbot (chatuse_binary), was modeled using a Random Forest classifier (Breiman, 2001). Logistic regression was considered as the primary classifier but rejected due to the large number of dummy variables required to represent ordinal predictors, which would introduce collinearity and reduce interpretability. Random Forest was selected over logistic regression for two reasons. First, it natively accommodates ordinal predictors without requiring dummy coding. Second, Random Forest implicitly captures interaction effects between predictors through recursive partitioning, allowing the hypothesized income × education compounding effect to emerge from the data without manual specification of an interaction term. All models used 100 trees with no maximum depth constraint and a fixed random seed (42) for reproducibility.

Encoding Strategy. A hybrid encoding approach was applied to the predictor variables. Education (educ_6cat, 6 levels), income (inc_9cat, 9 levels), and age (age_cat, 4 levels) were retained as raw integer codes, as their numeric ordering carries meaningful rank information and Random Forest splits respect this ordinality. Gender (3 categories) and race/ethnicity (5 categories) are nominal variables with no meaningful numeric ordering; treating them as integers would impose a false metric structure. These variables were therefore dummy-coded with one reference category dropped per variable to avoid perfect multicollinearity, yielding a final full feature matrix of nine predictors.

Model Comparison. Two nested classifier models were evaluated: a socioeconomic model including only education and income, and a full demographic model additionally incorporating age, gender, and race/ethnicity. Both models were evaluated using 5-fold cross-validation, with mean accuracy and standard deviation reported across folds. A naive majority-class baseline was computed from the marginal class distribution of chatuse_binary to provide a performance reference.

Secondary Outcome. Frequency of AI use (useai) is a 5-point ordinal scale ranging from 1 ("Almost constantly") to 5 ("Less often"). Because ordinal outcomes do not satisfy the equal-interval assumption underlying regression-based metrics such as MSE and R^2, a regression approach was not used for this outcome. Instead, Spearman rank correlations were computed between each socioeconomic predictor and useai. Spearman's ρ is appropriate here as it measures monotonic association without assuming interval scaling.

Visualization. To illustrate the income-by-education adoption gradient, the socioeconomic model was fit on the full analytic sample and used to generate predicted adoption probabilities across a complete grid of all 54 income × education combinations via predict_proba, displayed as an annotated heatmap. Feature importances were extracted from the full demographic model to quantify the relative contribution of each predictor.

3. Results

3a. Encoding Validation. Ordinal and dummy-encoded versions of the socioeconomic predictors were compared on a held-out test set prior to cross-validation; both yielded identical accuracy (0.588), confirming that ordinal encoding does not disadvantage the Random Forest classifier for these predictors.

3b. Model Accuracy. The naive majority-class baseline was 0.501, reflecting a near-perfectly balanced outcome variable. The socioeconomic model achieved a mean cross-validated accuracy of 0.590 (SD = 0.020), a modest but consistent improvement over baseline. The full demographic model achieved a mean accuracy of 0.623 (SD = 0.012), with the smaller standard deviation indicating greater stability across folds. The accuracy gain from adding demographic controls suggests that age, gender, and race/ethnicity contribute meaningful predictive information beyond socioeconomic status alone, though the overall modest performance of both models indicates that the variables measured here collectively explain only a limited portion of variation in AI adoption.

Table 1. 5-Fold Cross-Validation Accuracy

Model	Mean	Standard Deviation
Socioeconomic only	0.590	0.020
Full model	0.623	0.012
Naive Baseline	0.501	-

3c. Frequency of Use. Both Spearman correlations were statistically significant ($p < .001$). Education showed a weak negative association with useai ($\rho = -0.163$), and income showed a similarly weak negative association ($\rho = -0.125$). Because useai is reverse-coded, with lower scores indicating more frequent use, these negative correlations indicate that higher education and higher income are both associated with greater frequency of AI interaction. This is consistent with the socioeconomic gradient observed in the adoption analysis. Education emerged as the slightly stronger predictor ($\rho = -0.163$ vs. $\rho = -0.125$), though both correlations were modest in magnitude, suggesting that most variation in use frequency is driven by factors not captured in the present model.

Table 2. Spearman Correlations with AI Use Frequency (useai)

Education (educ_6cat)	rho (ρ) = -0.163	p = 0.0000
Income (inc_9cat)	rho (ρ) = -0.125	p = 0.0000

3d. Feature Importance. Contrary to the paper's hypothesis that education would emerge as the stronger socioeconomic predictor of adoption, income ranked highest in feature importance overall (0.322), followed by age (0.242) and education (0.229). Gender and race/ethnicity dummy indicators contributed substantially less, each falling below 0.055. The relative proximity of income and education importances suggests that both socioeconomic dimensions contribute comparably to predicting adoption, with neither clearly dominating once demographic controls are included. It is worth noting that income's apparent advantage over education in the classifier may partly reflect its finer granularity, with nine income categories versus six education levels, giving the model more decision boundaries to exploit.

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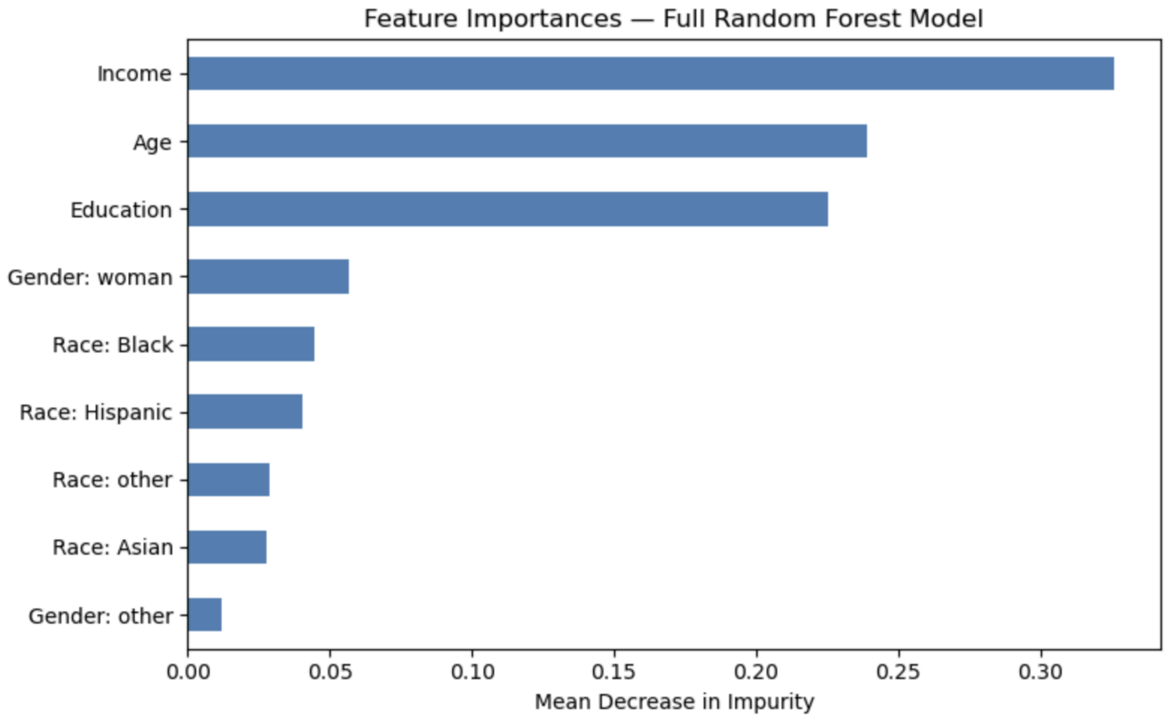

Figure 1. Feature importances from the full Random Forest model, showing each predictor's relative contribution to classification accuracy as measured by mean decrease in impurity.

3e. Adoption Probability Heatmap. Predicted adoption probabilities across the income × education grid broadly increased with both higher education and higher income, consistent with the hypothesized privilege gradient. However, the pattern was not uniformly monotonic. The less-than-high-school row exhibited notable instability, with predicted probabilities ranging from 0.00 to 0.77 across income levels This likely reflects sparse cell counts at low education levels rather than a true underlying pattern. Cells with higher education and higher income, particularly postgraduate respondents across most income brackets, showed more consistently elevated adoption probabilities, with several cells exceeding 0.60. Whether this reflects a true compounding effect beyond the independent contributions of each variable cannot be determined from predicted probabilities alone, as the heatmap does not isolate the interaction term from the main effects.

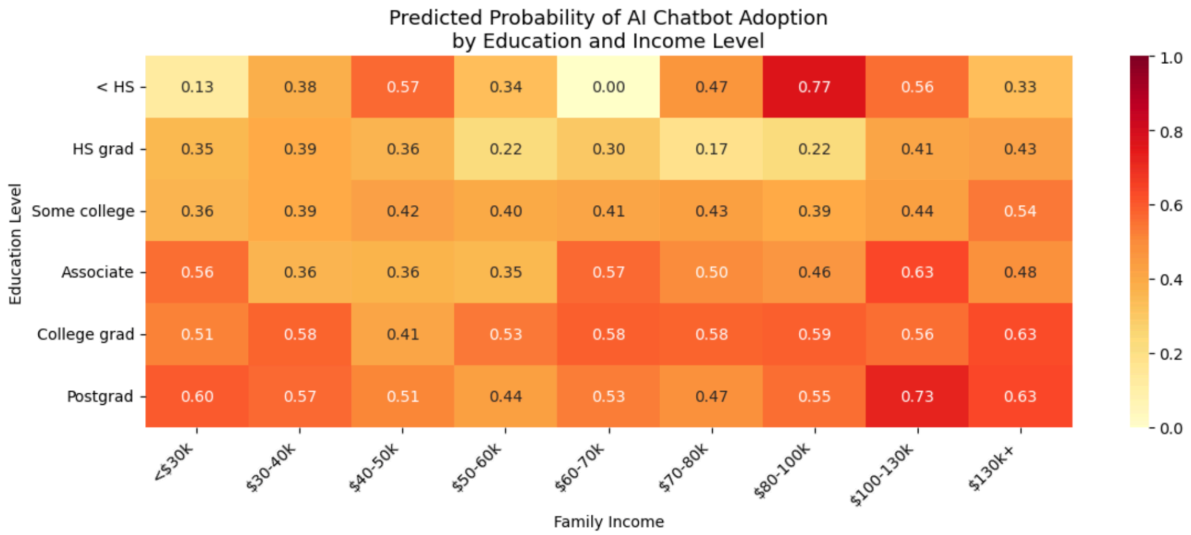

Figure 2. Predicted probability of AI chatbot adoption across education and income levels, estimated using the socioeconomic Random Forest model.

4. Discussion

4a. Findings. Both income and education were positively associated with AI chatbot adoption and more frequent use, supporting the premise that AI adoption follows a socioeconomic gradient. However, the gradient was weaker than expected, with classifier accuracy at 62.3% against a 50.1% baseline, suggesting that income and education alone explain only a modest portion of who adopts AI.

4b. Hypothesis Evaluation. The first hypothesis received partial support. Both socioeconomic predictors contributed to model performance and were significantly correlated with use frequency, but income (importance = 0.322) outranked education (0.229), contradicting the prediction that education would be stronger. One substantive explanation is that paid AI tiers create a direct cost barrier that income captures more immediately than education does. The second hypothesis, that high income and high education would interact to compound adoption, found tentative visual support in the heatmap. The pattern was too noisy to treat as confirmatory, particularly in sparse cells at low education levels. A partial dependence plot isolating the income × education interaction net of each variable's independent contribution would be needed to evaluate this hypothesis more rigorously.

4c. The Age Finding. Age (importance = 0.242) ranked second overall, nearly matching income, despite not being a focus of the original hypotheses. This suggests that generational familiarity with digital tools is at least as consequential as socioeconomic status in predicting adoption, and that policy discussions about the AI divide cannot focus on income and education alone.

4d. Lessons Learned. Three methodological takeaways stand out. First, the motivation for encoding choices should follow from measurement properties: dummy coding is unnecessary for ordinal predictors but remains necessary for nominal ones, and conflating the two would have been wrong in both directions. Second, evaluation metric choice matters. Accuracy was appropriate here given near-perfect class balance, but would be misleading under imbalance. Third, predicted probabilities from tree-based models in sparse feature-space regions are unreliable; the instability in the less-than-high-school heatmap row is a product of thin training data, not a real pattern.

4e. Limitations. The binary adoption outcome collapses meaningful variation in depth of use. Likely omitted predictors such as occupational field, digital literacy, and social network exposure to AI, probably account for much of the unexplained variance. The cross-sectional design precludes causal inference; whether income enables AI access or AI skill drives earnings cannot be determined from these data.

4f. Future Directions. Partial dependence plots would provide a cleaner test of the income × education interaction than the heatmap. Incorporating survey weights (WEIGHT_W152) would improve representativeness of descriptive results. Comparing Random Forest against gradient boosting would assess whether a more flexible model recovers additional signal. Most valuably, replicating this analysis across multiple ATP waves would reveal whether the socioeconomic adoption gap is widening or narrowing over time.

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